STA 301

Date:

Lec # 6

Types of avrages

- Arithmatic mean
- Geometric mean
- The Harmonic mean
- Mode

Arithmatic mean:-

Arithmatic mai Sub ka Sum krna hota Then usko Total py devid krna hota h.

Example:-

$$\bar{X} = \frac{1+2+3+4}{4} = \frac{10}{4} = 2.5$$

Arithmatic mean denoted by $\bar{X}$

Geometric mean

Variables ko multiply krna h aur unka root dena h. is mai 0 nhi ata agr a Jay To Ans 0 hoga.

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Examples

$$G.M = (1 \times 2 \times 3 \times 4)^{1/4}$$

$$= (24)^{1/4}$$

$$= 2.213$$

Harmonic mean

No of variables oper sakhmi hr aur baki no ka reciprocal lyna hr aur plus krna aur jo ans ay ga uss ko devid krna h No of Terms py jo oper hogi

$$H.M = \frac{4}{\frac{1}{1} + \frac{1}{2} + \frac{1}{3} + \frac{1}{4}}$$

$$= \frac{4}{1 + 0.5 + 0.33 + 0.25}$$

$$= \frac{4}{2.083}$$

$$= 1.92$$

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Mode

Most repeted value in your data.

it's denoted by

$\hat{X}$

1, 1, 2, 3, 4

Yaha Mode ? aygaa ?

$\hat{X} = 1$

Example:— Page (49)

Find Mode ?

Mileage Rating	class Boundary	No. of cars (f)	
30.0 — 32.9	29.95 — 32.95	2	
33.0 — 35.9	32.95 — 35.95	4	f_1
36.0 — 38.9	35.95 — 38.95	14	f_m
39.0 — 41.9	38.95 — 41.95	8	f_2
42.0 — 44.9	41.95 — 44.95	2	
		30	

$\Sigma f = 30$

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formulae

$$\hat{x} = l + \frac{(f_m - f_1)}{(f_m - f_1) + (f_m - f_2)} \times h \quad \text{---(1)}$$

$l = 35.95$, $f_m = 14$, $f_1 = 4$, $f_2 = 8$

putting the value in eq (1)

$$\hat{x} = 35.95 \frac{14-4}{(14-4)+(14-8)} \times 3$$

$$= 35.95 + \frac{10}{10+6} \times 3$$

$$= 35.95 + \frac{10}{16} \times 3$$

$$= 35.95 + \frac{30}{16}$$

$$= 35.95 + 1.875$$

$$= 37.825$$

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Lec #NO 7

Find Mean for Grouped data

$\frac{30.0+32.9}{2} = 31.45$ ←

Class Interval — Mileage Rating	frequency No of Cars (f)	Mid point (x)	f(x) fx
30.0 — 32.9	2	31.45	62.9
33.0 — 35.9	4	34.45	137.8
36.0 — 38.9	14	37.45	524.3
39.0 — 41.9	8	40.45	323.6
42.0 — 44.9	2	43.45	86.9
	$\Sigma f = 30$		$\Sigma fx = 1135.5$

$$\bar{X} = \frac{\Sigma fx}{\Sigma f}$$

$$= \frac{1135.5}{30}$$

Ug Group = 37.85

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Example:

find weighted Mean

X	W	WX
15	100	1500
5	80	400
10	70	700
2	50	100
	$\Sigma W = 300$	$\Sigma WX = 2700$

$$\bar{X}_w = \frac{\Sigma WX}{\Sigma W}$$

$$= \frac{2700}{300}$$

$$= 9$$

Un Median group ky liy jo data nikalty h na wo do terhan ka hota h

Case 1 odd

Case 2 even

Median?

for odd No

4, 2, 1, 3, 5

no of terms $n = 5$

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Arranged:

1, 2, 3, 4, 5

↓ middle no.

Median = 3

for even no

0, 1, 2, 5, 4, 7, 8, 9

Arranged

= 0, 1, 2, 4, 5, 7, 8, 9

$$= \frac{4+5}{2}$$

$$= \frac{9}{2}$$

$$= 4.5$$

⟶

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Lec # 8

find Median for grouped data:-

Mileage Rating	frequency No of cars	class Boundarys	cf
30.0 — 32.9	2	29.95 — 32.95	2 c
33.0 — 35.9	4	32.95 — 35.95	6
36.0 — 38.9	14 f	35.95 — 38.95	20
39.0 — 41.9	8	38.95 — 41.95	28
42.0 — 44.9	2	41.95 — 44.95	30
	$\Sigma f = 30$		

sub 2 (30.0 — 32.9)

sub = h (35.95 — 38.95)

Consider

$f = n$

$n = 30$

$\frac{n}{2} = \frac{30}{2}$

$n = 15$ see in table near digit

$$\tilde{x} = l + \frac{h}{f}\left(\frac{n}{2} - c\right) \quad \text{---------- (1)}$$

$l = 35.95$, $h = 3$

$f = 14$, $c = 6$

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putting the value in eq (1)

$$\tilde{X} = 35.95 + \frac{3}{14}(15-6)$$

$$= 35.95 + \frac{3}{14}(9)$$

$$= 35.95 + \frac{27}{14}$$

$$= 35.95 + 1.92$$

$$= 37.87$$

median $\tilde{X} = 37.9$

Next Topic

Quantiles Decline percentile

Quartiles jo h wo nikla h Quiter sy Quiter ko 4 hiso py devid kriy. To wo Desile py devide hota h.

Slide ky ander Quantile

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ky 4 points nhi honge.
jesy Quartile ka matlb 4
h to up 3 tuk jao
gy.

Mileage Rating	frequency No of cars	class Boundries	cf
30.0 — 32.9	2	29.95 — 32.95	2
33.0 — 35.9	4	32.95 — 35.95	6
36.0 — 38.9	14	35.95 — 38.95	20
39.0 — 41.9	8	38.95 — 41.95	28
42.0 — 44.9	2 f	41.95 — 44.95	30
	$\Sigma f = 30$		

first of all we see $f = n = 30$
Quarter h to 4 Py devide

$$\frac{n}{4} = \frac{30}{4} = \frac{15}{2} = 7.5$$

7.5 f mai kis ky qreb h
to ya h 8 cb yahn
Py 8 ka bar bany
gy.

MCQ $Q_2 = X$ median equal note h.

Date:

$$Q_1 = l + \frac{h}{f}\left(\frac{n}{4} - c\right) \quad \text{---(1)}$$

l = lower class boundry = 38.95

h = class interval = 3

f = frequency = 8

$\frac{n}{4}$ =

c = 20

Putting the value in eq (1)

Q_1 mai nky Sath 1xn, aur Q_2 mai nky 2xn

$$Q_2 = l + \frac{h}{f}\left(\frac{2 \times n}{2 \cancel{4}} - c\right)$$

MCQ $Q_2 = l + \frac{h}{f}\left(\frac{h}{2} - c\right) =$ eq (1)

$$Q_3 = l + \frac{h}{f}\left(\frac{3 \times n}{2} - c\right) \quad \text{---(iii)}$$

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Quartile / Decile Important Paper

Next am Talk about Decile

Decile nikla h 10 sy ab ya bat yad rakhin ky baky y devid ky nichy 10 aur h.

$$D_1 = l + \frac{h}{f}\left(\frac{n}{10} - c\right) \quad \text{---} \quad (I)$$

$$D_2 = l + \frac{h}{f}\left(\frac{2n}{10} - c\right)$$

$$D_2 = l + \frac{h}{f}\left(\frac{n}{5} - c\right) \quad \text{---} \quad (II)$$

$$D_3 = l + \frac{h}{f}\left(\frac{3 \times n}{10} - c\right) \quad \text{---} \quad (III)$$

Persentil % persent sy nikla h. matlb 100 sy ab baky ky nichy 100 aay gy.

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Date:

$$P_1 = \frac{l+h}{f}\left(\frac{n}{100} - c\right)$$

$$P_2 = l + \frac{h}{f}\left(\frac{2 \times n}{100} - c\right)$$

$$P_3 = l + \frac{h}{f}\left(\frac{3 \times n}{100} - c\right) \ldots$$

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STA 301 V-Imp

Lec #9

Geometric Mean engroup data?

engroup data jis mai frequency nhi di jati, table nhi diya jaty simple X aur y diy jaty h.

ager value choti h aur kam h. Then Simple formula lagyn gyn

$$G = \sqrt[n]{1 \times 2 \times 3}$$

$$G = (6)^{1/3}$$

Agr value bari hon G q value ho digits bary hon tasy 45.

Phr hum y formula use krain gyn.

$$G.M = \text{Anti log}\left[\frac{\sum \text{Log } X}{n}\right]$$

Sum

Example:-

Given

x	Log x
45	1.6532
32	1.5052
37	1.5682
46	1.6628
39	1.5911
36	1.5563
41	1.6128
48	1.6812
36	1.5563
	$\Sigma \log x = 14.3870$

As we know that

$$G.M = \text{Antilog}\left[\frac{\Sigma \log x}{n}\right]$$

putting the value

$$n = 9, \quad \Sigma \log x = 14.3870$$

$$G.M = \text{Antilog}\left[\frac{14.3870}{9}\right]$$

$$= \text{Antilog}\,[1.5986]$$

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Shri log = = 39.68

Jo same hoti h wo class boundry hoti h Jo same nhi hoti wo class interval hoti h.

Midpoint ko x bolty h. x kahan lazmi h.

first of all we see formula:-

$$G.M = \text{Anilog}\left[\frac{\Sigma f \log x}{\Sigma f}\right]$$

Example:-

Mileage Rating	No of cars f	class Mark Mid point x	log x	f log x
30.0 — 32.9	4	31.45	1.4976	2.9952
33.0 — 35.9	14	34.45	1.5372	6.1488
36.0 — 38.9	8	37.45	1.5735	22.0290
39.0 — 41.9	8	40.45	1.6069	12.8552
42.0 — 44.9	2	43.45	1.6380	3.2760
	$\Sigma f = 30$			
				$\Sigma f \log x = 47.3042$

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Date:

As we know that

$$G.M = \text{Antilog}\left[\frac{\Sigma f \log x}{\Sigma f}\right]$$

Putting value

$$G.M = \text{Antilog}\left[\frac{47.3040}{30}\right]$$

$$\text{Antilog}(1.5768)$$

$$G.M = 37.74$$

Important Point

Find percentage average GM nikalna h.

km per hour

Speed average key liy H.M nikalna h.

percentage average increase kis sall mai ho rahi h.

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Sol:-

year	Turv	Persentage compared with year Earlier
1958	£ 2,000	—
1959	£ 2,500	125
1960	£ 5,000	200
1961	£ 7,500	150
1962	£ 10,500	140

G.M = $\sqrt[4]{125 \times 200 \times 150 \times 140}$

$= (525000000)^{1/4}$

$= 151.37\%$

So,

$151.37\% \times 200 = 3027$

$151.37\% \times 3027 = 4583$

$151.37\% \times 4583 = 6937$

$151.37\% \times 6937 = 10500$

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Date:

find average = Mean?
Tu osky liy = H.M lagayn gy.

Harmonic Mean:
for ungroup data

$$H.M = \frac{n}{\Sigma\left(\frac{1}{n}\right)}$$

for grouped data:

$$H.M = \frac{f}{\Sigma f(1/x)}$$

Example:=

x	1/x
30	1/30
35	1/35
40	1/40
40	1/40
45	1/45
40	1/40
50	1/50
55	1/55
55	1/55
30	1/30

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$$H.M = \frac{n}{\Sigma(\frac{1}{n})}$$

$n = 10$

$\Sigma(1/n) = 0.2988$

putting value

$$H.M = \frac{10}{0.2988}$$

$$42.2 \text{ m}$$

$\longleftrightarrow$

Average of 10 student out of 20

$$\frac{13+11+9+9+6+5+19+17+12+9}{10}$$

$$\frac{110}{10} = 11$$

Average of 20 student

$$= \frac{11}{20}$$

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Lec #9

Relation b/w $\bar{X}$, GM, H.M

$$\bar{X} \geq GM \geq H.M$$

Range formula = $X_m + X_0$

Mid Range = $\frac{X_m + X_0}{2}$

Example:-

1, 2, 10

$$= \frac{10+1}{2}$$

$$= \frac{11}{2} = 5.5$$

Mid Quatile Range

$$\frac{Q_3 + Q_1}{2}$$

Example:- 4, 5, 2

$$= \frac{2+4}{2}$$

$$= 6/2$$

$$= 3$$

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S.I A

Lec' #10

Find combine Avorage ?
Area A and Area B
Given Data

No of Pupils	No of Area A	Area B	Midpoint $\bar{x}$
10–14 $\frac{10+14}{2}=12$	0	55	12
15–19	3	8	17
20–24	13	10	22
25–29	24	12	27
30–34	17	14	32
35–39	3	5	37
40–44	0	3	41
45–49	0	3	47
	$\Sigma A=60$	$\Sigma B=60$	$\Sigma \bar{x}=235$

Avorage Area of A $= \bar{X}_A = \frac{\Sigma A}{n}$

$= \frac{60}{8} \Rightarrow 7.5$

Avrage Area of B $= \bar{X}_B = \frac{\Sigma B}{n}$

$\frac{60}{8} = 7.5$

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$$\bar{X} = \frac{235}{8}$$

$\bar{X} = 29.375$ mistake on book

coefficient of dispersion =

$$= \frac{x_m - x_0}{x_m + x_0}$$

Example:-

10, 0, 20, 300

$x_m = 30, \quad x_0 = 0$

coefficient of dispersion

$$= \frac{300 - 0}{300 + 0}$$

$$= \frac{300}{300}$$

$$= 1$$

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Quadile Deviation:-

$$QD = \frac{Q_3 - Q_1}{2}$$

Quartile	company X	company Y
1 Quartile	60 share	165 shares
2 Quartile	185	185
3 Quartile	270	210

for x :-

$$Q.D = \frac{Q_3 - Q_1}{2}$$

$$= \frac{270 - 60}{2} = 105$$

for y :-

$$\frac{Q_3 - Q_1}{2} = \frac{210 - 165}{2}$$

$$= 22$$

Coefficient Quartile Deviation

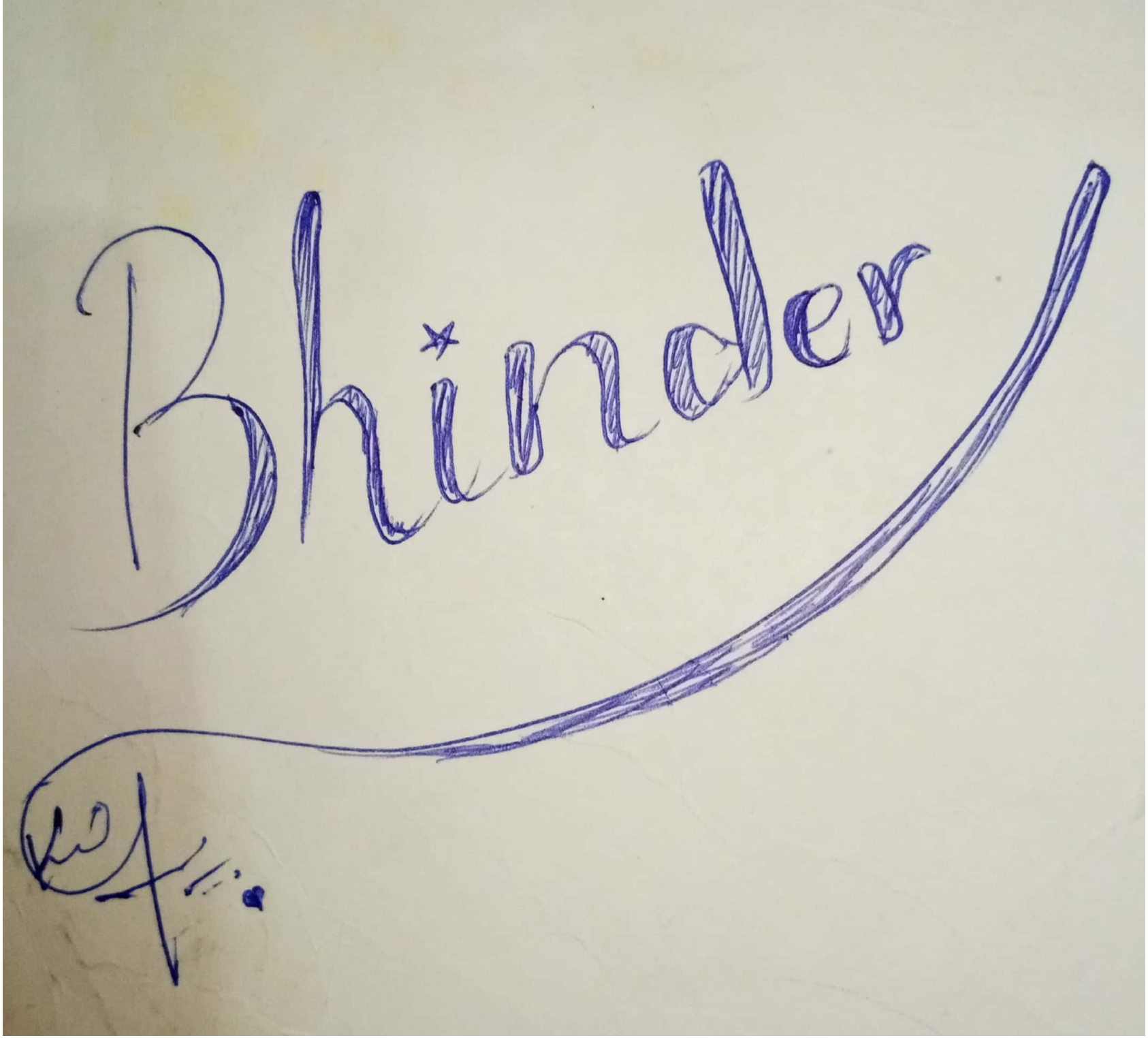
$$= \frac{Q_3 - Q_1}{Q_3 + Q_1}$$

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Mean deviation =

$$M.D = \frac{\Sigma |d|}{n}$$

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